Corresponding parts of congruent triangles

1

a $\angle AZP$ b $\angle ZAP$ c $\angle ZPA$

2

a No b Yes c Yes d Yes

3

a

i $\triangle ABC \cong \triangle EFD$

ii SAS: A pair of corresponding sides and the included angle are congruent.

iii $\angle EDF$

b

i $\triangle ABC \cong \triangle OPQ$

ii SSS: All three corresponding sides are congruent.

iii $\angle OQP$

4 68°

5

a $\triangle QRS \cong \triangle GHF$ b Side-side-side congruence (SSS)

c $g = 64$

6

a Yes b Yes c No d No

e No

7

a $\triangle EDF \cong \triangle XYZ$ b Side-angle-side congruence (SAS)

c $y = 5$

8

a Side-angle-side congruence (SAS) b $p = 37$

9

To prove: [redacted] $\cong \angle XZY$

	Statements	Reasons
1.	$\angle CAB \cong \angle ZXY$	Given
2.	$\angle CBA \cong \angle ZYX$	Given
3.	$\overline{AB} \cong \overline{XY}$	Given
4.	$\triangle ABC \cong \triangle XYZ$	Angle-side-angle congruence (ASA)
5.	$\angle ACB \cong \angle XZY$	Corresponding parts of congruent triangles are congruent (CPCTC)

10

To prove: $\angle ACB \cong \angle XZY$

	Statements	Reasons
1.	$\overline{CB} \cong \overline{ZY}$	Given
2.	$\overline{AC} \cong \overline{XZ}$	Given
3.	$\overline{AB} \cong \overline{XY}$	Given
4.	$\triangle ABC \cong \triangle XYZ$	Side-side-side congruence (SSS)
5.	$\angle ACB \cong \angle XZY$	Corresponding parts of congruent triangles are congruent (CPCTC)

11

To prove: $\overline{SR} \cong \overline{PQ}$

	Statements	Reasons
1.	$\angle PSR \cong \angle RQP$	Given
2.	$\overline{PQ} \parallel \overline{SR}$	Given
3.	$\angle SRP \cong \angle QPR$	Alternate interior angles theorem
4.	$\overline{PR} \cong \overline{RP}$	Reflexive property of congruence
5.	$\triangle PQR \cong \triangle RSP$	Angle-angle-side congruence (AAS)
6.	$\overline{SP} \cong \overline{RQ}$	Corresponding parts of congruent triangles are congruent (CPCTC)

12

To prov[redacted] $\cong \overline{FG}$

	Statements	Reasons
1.	E is the midpoint of $\overline{DF}$	Given
2.	$\angle EDH \cong \angle EFG$	Given
3.	$\angle DHE \cong \angle FGE$	Given
4.	$\overline{DE} \cong \overline{EF}$	Properties of a midpoint
5.	$\triangle DEH \cong \triangle FEG$	Angle-angle-side congruence (AAS)
6.	$\overline{DH} \cong \overline{FG}$	Corresponding parts of congruent triangles are congruent (CPCTC)

13

To prove: $\triangle RMP$ is isosceles

	Statements	Reasons
1.	$\overline{RQ} \cong \overline{QP}$	Given
2.	$\overline{RP} \perp \overline{MQ}$	Given
3.	$\angle MQR$ and $\angle MQP$ are right angles	Definition of perpendicular
4.	$\angle MQR \cong \angle MQP$	All right angles are congruent
5.	$\overline{MQ} \cong \overline{MQ}$	Reflexive property of congruence
6.	$\triangle RMQ \cong \triangle PMQ$	Side-angle-side congruence (SAS)
7.	$\overline{RM} \cong \overline{PM}$	Corresponding parts of congruent triangles are congruent (CPCTC)
8.	$\triangle RMP$ is isosceles	Isosceles triangles are triangles with two congruent sides

14

a Yes

b No

c No

d Yes

Additional questions

15

a $\overline{DE}$

b $\overline{XZ}$

16

a $\angle N$ **b** $\angle A$ **c** b

17

a $\triangle BDE$

b

i $\angle BED$ **ii** $\angle EBD$

c

i DE **ii** DB

18

a 32 **b** 7

19

a 4 **b** 6 **c** 57° **d** 25°

20

a Answers may vary

$\Delta ABC \cong \Delta DEF$ and $\Delta ABC \cong \Delta FED$

b This answer will depend on the order of the congruency statement in part (a).

For $\Delta ABC \cong \Delta DEF$

$\overline{AB} \cong \overline{DE}$	$\angle A \cong \angle D$
$\overline{BC} \cong \overline{EF}$	$\angle B \cong \angle E$
$\overline{DF} \cong \overline{AC}$	$\angle C \cong \angle F$

For $\Delta ABC \cong \Delta FED$

$\overline{AB} \cong \overline{FE}$	$\angle A \cong \angle F$
$\overline{BC} \cong \overline{ED}$	$\angle B \cong \angle E$
$\overline{AC} \cong \overline{FD}$	$\angle C \cong \angle D$

c The order of the congruence statement should indicate the correspondency of the sides and angles. This statement tells us what angles and what sides are congruent from the corresponding parts of congruent triangle are congruent theorem even without a picture. Writing $\Delta ABC \cong \Delta DEF$ instead of $\Delta ABC \cong \Delta FED$ gives us different pairs of congruent sides and angles.

21

a

i Side-side-side congruency theorem (SSS)

ii Answers will vary.
$\Delta CDE \cong \Delta LMN$

iii $\angle D \cong \angle M$
$\angle C \cong \angle L$
$\angle E \cong \angle N$

b

i Angle-side-angle congruency theorem (ASA)

ii Answers will vary.
$\Delta PQR \cong \Delta STU$

iii $\angle T \cong \angle Q$
$\overline{PQ} \cong \overline{ST}$
$\overline{QR} \cong \overline{TU}$

c

i Angle-angle-side congruency theorem (AAS)

ii Answers will vary.
$\Delta GHI \cong \Delta LMN$

iii $\angle H \cong \angle M$
$\overline{GI} \cong \overline{LN}$
$\overline{GH} \cong \overline{LM}$

d

i Side-angle-side congruency theorem (SAS)

ii Answers will vary.
$\Delta ABC \cong \Delta DEF$

iii $\angle B \cong \angle E$
$\angle C \cong \angle F$
$\overline{BC} \cong \overline{EF}$

22

a Side-side-side congruency theorem (SSS)
$q = 67$

b Side-angle-side congruency theorem (SAS)
$p = 37$

c Angle-angle-side congruency theorem (AAS)
$x = 147$

d Angle-angle-side congruency theorem (AAS)
$y = 2$

23 We are given that $\angle A \cong \angle X$, $\angle B \cong \angle Y$, and $\overline{AB} \cong \overline{XY}$. Therefore, $\Delta ABC \cong \Delta XYZ$ by Angle-side-angle congruency theorem (ASA), and so $\overline{BC} \cong \overline{YZ}$ because corresponding parts of congruent triangles are congruent (CPCTC).

24

$\overleftrightarrow{PQ} \parallel$ [redacted]
Given

$\angle S \cong \angle Q$
Given

$\overline{PR} \cong \overline{RP}$
Reflexive property of equality

$\angle PRS \cong \angle RPQ$
Alternate Interior Angles Theorem

$\triangle PQR \cong \triangle RSP$
AAS

$\overline{PS} \cong \overline{QR}$
CPCTC

25 We are given that F is the midpoint of $\overline{DE}$ which means that $\overline{DF} \cong \overline{FE}$ by definition of midpoint. Then, we know that $\overline{DH} \parallel \overline{FG}$ so by the corresponding angles postulate $\angle HDF \cong \angle GFE$. Next, we are given that $\overline{DH} \cong \overline{FG}$, therefore $\Delta DFH \cong \Delta FEG$ by the Side-angle-side congruency theorem (SAS). Now we can conclude that $\angle DFH \cong \angle FEG$ because corresponding parts of congruent triangles are congruent (CPCTC). Finally, $\overline{FH} \parallel \overline{EG}$ by the converse of corresponding angles theorem.

26

To prove: $\Delta PQX \cong \Delta RQX$

	Statements	Reasons
1.	$\overline{PQ} \cong \overline{RQ}$ and $\overline{PS} \cong \overline{RS}$	Given
2.	$\overline{QS} \cong \overline{QS}$	Reflexive property
3.	$\Delta PQS \cong \Delta RQS$	Side-side-side congruency theorem (SSS)
4.	$\angle PQX \cong \angle RQX$	Corresponding parts of congruent triangles are congruent (CPCTC)
5.	$\overline{QX} \cong \overline{QX}$	Reflexive property
6.	$\Delta PQX \cong \Delta RQX$	Side-angle-side congruence theorem (SAS)